

Smart Power Optimizer

Single Input 1 For 1



PLC Transceiver

Single Input 1 For 2



Smart Logger



SafeSolar

Product Features



Module-level Power Optimization

Solve power generation losses caused by shading, different orientation and mismatching



Module-level Data Monitoring

Snapshots Include Voltage, current, power, power generation, IV working point, temperature, etc.



Module-level Rapid Shutdown

Automatic or manual shutdown, to ensure maintenance and fire safety



Bidirectional PLC Technology

Supports super-long communication distance



Flexible Configuration

Optimized long string design for reduce Installation costs



Fault Notification

Real-time fault information on abnormal situations of PV panels

Model	FR-PVMB-C65C	FR-PVMB-C80H	FR-PVMB-C130D	FR-PVMB-C130H
Number of Module	1 For 1	1 For 1	1 For 2	1 For 2
Max. Input/Output Power	650 W	800 W	1300 W	1300 W
Max. Input Voltage	70 Vdc	70 Vdc	125 Vdc	125 Vdc
Max. Continuous Input/Output Current	15 Adc	20 Adc	16 Adc	20 Adc
Dimension （ Without Cable & Connectors ）	132 × 120 × 25 mm	132 × 120 × 45 mm		
Operating Ambient Temperature Range	-40℃ ~ +85℃ ¹			
Installation Part	PV Rail ²			
Weight	≤ 750 g			
IP Rating	IP68			
Max. System Voltage	1500 Vdc (IEC/UL Standard)			
Max. Cable Loop Length	1600 m			
Communication Protocol	PLC （ Power Line Commnication）			
Cable Specs & Connectors Type	4.0 mm² Cables + MC4 Compatible ³			
Safety Compliance	NEC 2017, 2020 & 2023 (690.12); IEC 62109-1; EN 62109-1			
Standard Features				
Module-level Optimization	√	√	√	√
Advanced Features				
Automatic Rapid Shutdown	√	√	√	√
Manual Rapid Shutdown	√	√	√	√
Monitoring Platform				
Module-level Asset Management	√	√	√	√
2D & 3D Module-level Data Display	√	√	√	√
Multi-Module Data Comparison Analysis	√	√	√	√
Real-time Status Display	√	√	√	√
Physical Plants Overview	√	√	√	√
Energy Management	√	√	√	√
Data Dashboard	√	√	√	√

*1 When the ambient temperature reaches between 70°C to 85°C, it might operate under the over-temperature protection.

*2 The product is designed exclusively for installation on PV rail and should not be installed on the module frame.

*3 The cable specifications and length can be customized.

